

Nitin Lohan Ph.D.

AI Trainer & LLM Evaluator | RLHF & SciCode Benchmark Developer | Applied Mathematics Ph.D.

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PROFESSIONAL SUMMARY

Applied Mathematics Ph.D. and AI Trainer with hands-on experience at Mercor, Turing, micro1 and Handshake AI, building and evaluating training and benchmark data for large language models. Specializes in RLHF and reward-model preference data, SciCode-style scientific-coding benchmarks, inverse-problem and numerical-methods tasks, rubric design and model-failure analysis. Delivers reproducible Python code, test cases and reference solutions through GitHub-based workflows. Deep domain expertise in numerical PDEs, extreme-value statistics, weather modeling (WRF/WRFDA) and HPC, with 8 peer-reviewed journal articles; currently a National Post-Doctoral Fellow at NIT Rourkela working on numerical and machine-learning forecasting.

CORE SKILLS

RLHF & LLM Evaluation	RLHF / human-feedback data; reward-model preference ranking; LLM response grading; rubric and annotation-guideline design; red-teaming and model-breaking reasoning tasks; hallucination, error and edge-case detection; reference-solution writing; inter-rater consistency; citation and numerical verification; dataset quality assurance
Benchmark & Task Design	SciCode-style multi-step scientific-coding benchmarks; STEM benchmark development; inverse-problem tasks; exact-match and rubric-based scoring; discriminating test-case engineering; operator and scoring-logic redesign; evaluation-harness debugging; reproducibility validation
Mathematics & Statistics	Inverse problems; numerical linear algebra; partial differential equations; numerical analysis and interpolation (incl. WENO, monotone Hermite); finite-difference and finite-element methods; optimization; probability, statistics and extreme-value theory; change-point tests
Programming & ML	Python (expert): NumPy, SciPy, pandas, xarray, Matplotlib, Cartopy, wrf-python; machine learning for forecasting; Jupyter notebooks; reproducible, research-grade scientific code
Scientific Computing	WRF / WRFDA; 3DVAR data assimilation; HPC (SLURM/PBS); parallel computing; automation and scheduling; CDO; ERA5, IMDAA, IMD, GPM and INSAT datasets
Engineering Tools	Git/GitHub (repository-based task delivery); LaTeX; JupyterLab; VS Code; Cursor; Claude Code; Linux; Windows

PROFESSIONAL EXPERIENCE

AI Trainer / Evaluator & STEM Benchmark Developer Sep 2025 – Present
Mercor · Turing · micro1 · Handshake AI (Remote, contract)

- Produced RLHF and reward-model preference data: ranked, graded and critiqued LLM responses in mathematics and scientific computing against detailed rubrics.
- Designed 100+ high-difficulty mathematics and numerical-methods benchmark tasks to evaluate and improve advanced LLM reasoning.
- Built SciCode-style multi-step Python benchmarks delivered through GitHub repositories, with reference implementations, discriminating test cases and documentation, including WENO-type reconstruction tasks tied to numerical weather prediction.
- Created inverse-problem tasks (recovering unknown parameters or sources from observed data) that test model reasoning on ill-posedness, stability and regularization.
- Authored rigorous, step-by-step reference solutions with explicit assumptions, derivations, unit consistency and verified final answers.
- Evaluated model outputs across multiple model families for mathematical correctness, reasoning depth, hallucinations, ambiguity, formatting and answer consistency.
- Revised tasks through 3–4 reviewer cycles, redesigning operators and scoring logic for exact-match evaluation; verified every numerical output and citation against source PDFs before delivery, reducing reviewer rejections.

National Post-Doctoral Fellow (NPDF)

Oct 2026 – Present

Department of Earth and Atmospheric Sciences, NIT Rourkela · Odisha

- Research on numerical (NWP) and machine-learning forecasting over the Himalayan region.

Assistant Professor (Mathematics)

Jul 2025 – Sep 2026

G.L. Bajaj Institute of Technology and Management · Greater Noida

- Taught engineering mathematics to 100+ undergraduates; developed course material, assessments and problem sets.
- Mentored students in applied mathematics, numerical methods and scientific computing with Python.

Doctoral Researcher (Applied Mathematics)

Jul 2020 – Jan 2026

Gautam Buddha University · Greater Noida

- Diagnostic and numerical-modeling studies of heavy-precipitation and cloudburst events over India using ERA5, IMDAA, IMD, GPM and INSAT data.
- Ran WRF/WRFDA simulations on HPC clusters with 3DVAR data assimilation for high-impact case studies.

SELECTED PROJECTS

STEM Benchmark Development for LLM Evaluation · Mercor / Turing

- Developed challenging mathematics and numerical-analysis tasks across 15+ benchmark projects, exposing model weaknesses in reasoning, derivation and numerical consistency.

SciCode-Style Multi-Step Python Benchmarks · GitHub-based delivery

- Built 25+ reproducible benchmark notebooks spanning FEM, phase-field fracture, poroelasticity and stochastic-flow problems from recent arXiv papers.
- Diagnosed harness artifacts versus genuine model failures and recalibrated evaluation pipelines for fair scoring.

Statistics of Monsoon Rainfall Extremes, 1951–2025 · Atmosphere (2026)

- Analyzed 75 years of 0.25° IMD gridded daily rainfall in Python; fitted Brown–Resnick extremal-dependence models to F-madogram estimates and applied sliding-window trend and Pettitt change-point tests.

EDUCATION

Ph.D. in Applied Mathematics, Gautam Buddha University, Greater Noida

2020 – 2026

Thesis: *Diagnostic and Numerical Modeling Study of Heavy Precipitation Events over Indian Region* ·

Supervisor: Dr. Sushil Kumar · CGPA 8.29

M.Sc. in Mathematics, DAV College, Panjab University, Chandigarh

2016 – 2018

B.Sc. (Physics, Chemistry, Mathematics), Govt. PG College Jind, Kurukshetra University

2013 – 2016

SELECTED PUBLICATIONS

- Lohan, N., Kumar, S., Ben Hasher, F. F., & Zhran, M. (2026). Reorganization of the spatial dependence structure of Indian summer monsoon rainfall extremes, 1951–2025. *Atmosphere*, 17(9), 892. doi:10.3390/atmos17090892
- Lohan, N., Pattanaik, D. R., Kumar, S., & Giri, R. K. (2025). Diagnostic study of monsoon-linked rainfall events: Unraveling the role of monsoon low pressure systems across the Indian subcontinent. *Earth Systems and Environment*. doi:10.1007/s41748-025-00706-y
- Lohan, N., Routray, A., Kumar, R., Mahala, B. K., & Kumar, S. (2025). Validation and diagnostic study of cloudburst events over the Himalayan region using IMDAA and ERA5. *Geomatics, Natural Hazards and Risk*, 16(1), 2446589. doi:10.1080/19475705.2024.2446589

Full list (8 journal articles, 4 conference papers / book chapters): [Google Scholar](#)

QUALIFICATIONS & TRAINING

- CSIR-UGC NET (Mathematical Sciences), qualified 2019 and 2022 · GATE Mathematics, qualified 2022 · Languages: Hindi (native), English (fluent)
- Research collaborations with IMD, NCMRWF and IITM Pune on HPC-based modeling and 3DVAR data assimilation · Training: IMD-Pune Induction Training Program on Meteorology (2024); NIDM Disaster Risk Reduction Programme (2023)